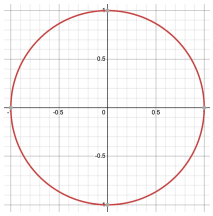
1 Parametric Curves

A circle $x^2+y^2=R^2$ is not the graph of a function $y=f(x)$ since it fails the vertical line test.



However, we can describe it as a function by taking x and y to both be functions of a third variable t, for example by taking

$$(x,y)=(R\cos(t),R\sin(t)) \quad 0\le t\le 2\pi.$$

Thinking of t as time, as t increases from 0 to 2π, the circle is traced out once counterclockwise, starting from the point $(R,0)$. Note that the input of this function is a real number, while the output is a vector in $\mathbb{R}^2$. In this context, t is called the *parameter*, the equations $x=f(t)$, $y=g(t)$ are called *parametric equations*, and the image of the function $(x,y)=(f(t),g(t))$ is called a *parametric curve*.

1. Sketch the following parametric curves by creating a table with columns $(t,x(t),y(t))$. Evaluate $(x(t),y(t))$ for integer values of t and connect the dots. Viewing t as time, indicate the orientation along which the curve is traced out by drawing an arrow:

   (a) $x(t)=t-1, y(t)=2t+4$, $-3\le t\le 2$

   (b) $x(t)=t^2-3$, $y(t)=2t+1$, $-2\le t\le 3$

One way to better understand a curve given by parametric equations $(x,y) = (f(t), g(t))$ is to eliminate the parameter t and rewrite the parametric equations as a single equation relating the variables x and y.

For example, consider the parametric curve defined by $x(t) = t^2 - 3$, $y(t) = 2t + 1$ in 1.(b) above. We see that $t = \frac{y-1}{2}$, so

$$x = \left(\frac{y-1}{2}\right)^2 - 3 = \frac{y^2 - 2y - 11}{4},$$

from which we see that the curve is a parabola opening to the right.

2. Eliminate the parameter for each of the plane curves below and rewrite the parametric equations as a single equation relating x to y.

   (a) $x(t) = \sqrt{2t+4}$, $y(t) = 2t + 1$, $-2 \leq t \leq 6$

   (b) $x(t) = 4\cos(t)$, $y(t) = 3\sin(t)$, $0 \leq t \leq 2\pi$

We have just seen how to eliminate the parameter from a set of known parametric equations. Suppose instead we are given an equation relating x and y. How can we rewrite this as a pair of equations in parametric form? This is called *parametrization*. In general, there may be many different ways of doing so. If $y = f(x)$ defines y as a function of x, then the easiest way is to just view x as the parameter:

$$x(t) = t, \quad y(t) = f(t).$$

For example, if we can parametrize $y = 3x^2 - 1$ as

$$x(t) = t, \quad y(t) = 3t^2 - 1.$$

3. Parametrize the curve $y = x^3 e^x$.

2 Calculus of Parametric Curves

2.1 Tangents

How can we find the slope of the tangent line to a given point of a parametric curve? That is, if we are given parametric functions $x = f(t)$, $y = g(t)$, how can we find $\frac{dy}{dx}$? One way is to eliminate the parameter t, and then use implicit differentiation. The next theorem allows us to find $\frac{dy}{dx}$ without the need to eliminate t.

Theorem 2.1 (Slope of the tangent line to parametric curve)**.** *Consider the plane curve defined by the parametric equations $x = x(t)$ and $y = y(t)$. At a point where y is a differentiable function of x, if $x'(t)$ and $y'(t)$ both exist and $x'(t) \neq 0$, then*

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)}.$$

4. Prove Theorem 2.1. (*Hint*: Use the Chain Rule.)

5. Prove that the second derivative is given by the formula

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}.$$

(*Hint*: Apply Theorem 2.1 to the parametric equations $x = x(t)$, $\frac{dy}{dx} = \frac{dy}{dx}(t)$.)

6. A curve C is defined by parametric equations $x = t^2$, $y = t^3 - 3t$.

   (a) Show that C has two tangents at the point $(3,0)$ and find their equations.

   (b) Find the points on C where the tangent is horizontal or vertical.

   (c) Determine where the curve is concave upward or downward.

   (d) Sketch the curve.

3 Arc Length

We learned previously how to find the length L of a curve C from $a \leq x \leq b$ when C is the graph of a function $y = F(x)$ with F' continuous:

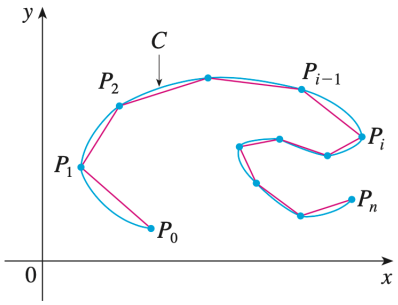
$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

Suppose now that C can also be described by the parametric equations $x = f(t)$ and $y = g(t)$, $\alpha \leq t \leq \beta$, where $\frac{dx}{dt} > 0$ (this condition ensures that C is traversed once, from left to right, as t increases from α to β).

7. Use the formula for $\frac{dy}{dx}$ in Theorem 2.1 and the substitution rule to show that we can rewrite L as

$$L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt. \tag{3.1}$$

Even if C can't be expressed in the form $y = F(x)$, the formula in Equation (3.1) is still valid. We will now sketch this argument. Divide the parameter interval $[\alpha, \beta]$ into n subintervals of equal width Δt. If $t_0, t_1, t_2, \ldots, t_n$ are the endpoints of these subintervals, then $x_i = f(t_i)$ and $y_i = g(t_i)$ are the coordinates of points $P_i(x_i, y_i)$ that lie on C and the polygon with vertices $P_0, P_1, \ldots, P_n$ approximates C.



We define the length L of$C$ to be the limit of the lengths of these approximating polygons as $n \to \infty$:

$$L = \lim_{n\to\infty} \sum_{n=1}^{\infty} |P_{i-1}P_i|.$$

The Mean Value Theorem, when applied to f on the interval $[t_{i-1}, t_i]$, gives a number t_i^* in (t_{i-1}, t_i) such that

$$f(t_i) - f(t_{i-1}) = f'(t_i^*)(t_i - t_{i-1}).$$

If we let $\Delta x_i = x_i - x_{i-1}$ and $\Delta y_i = y_i - y_{i-1}$, this equation becomes

$$\Delta x_i = f'(t_i^*)\Delta t.$$

Similarly, when applied to g, the Mean Value Theorem gives a number t_i^{**} in (t_{i-1}, t_i) such that

$$\Delta y_i = g'(t_i^{**})\Delta t.$$

Therefore

$$\begin{aligned} |P_{i-1}P_i| &= \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} = \sqrt{[f'(t_i^*)\Delta t]^2 + [g'(t_i^{**})\Delta t]^2} \\ &= \sqrt{[f'(t_i^*)]^2 + [g'(t_i^{**})]^2}\Delta t \end{aligned}$$

and so

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{[f'(t_i^*)]^2 + [g'(t_i^{**})]^2}\Delta t.$$

This sum resembles a Riemann sum for the function $\sqrt{[f'(t)]^2 + [g'(t)]^2}$ but it is not exactly a Riemann sum because $t_i^* \neq t_i^{**}$ in general. Nevertheless, if f' and g' are continuous, it can be shown that the limit above is the same as if t_i^* and t_i^{**} were equal, namely,

$$L = \int_{\alpha}^{\beta} \sqrt{[f'(t)]^2 + [g'(t)]^2}\,dt.$$

Thus, using Leibniz notation, we have the following result.

Theorem 3.2. *If a curve C is described by the parametric equations $x = f(t)$, $y = g(t)$, $\alpha \le t \le \beta$, where f' and g' are continuous on $[\alpha, \beta]$ and C is traversed exactly once as t increases from α to β, then the length of C is*

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt.$$

8. Use the arc length formula above to show that the circumference of the unit circle

$$x = \cos(t), \quad y = \sin(t), \quad 0 \le t \le 2\pi$$

is 2π.